

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C05: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Technical Presentation

Code of Conduct:

I G Amrutha(192511250) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

amrutha

Technical Presentation

- 1. Reactive Agent-Based Traffic Control System**
Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.
- 2. Goal-Based Agent for Healthcare Diagnosis**
Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.
- 3. Utility-Based Agent for E-Commerce Recommendations**
Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.
- 4. Learning Agent for Personalized Education System**
Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.
- 5. Multi-Agent System for Disaster Management**
Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Recommended presentation title

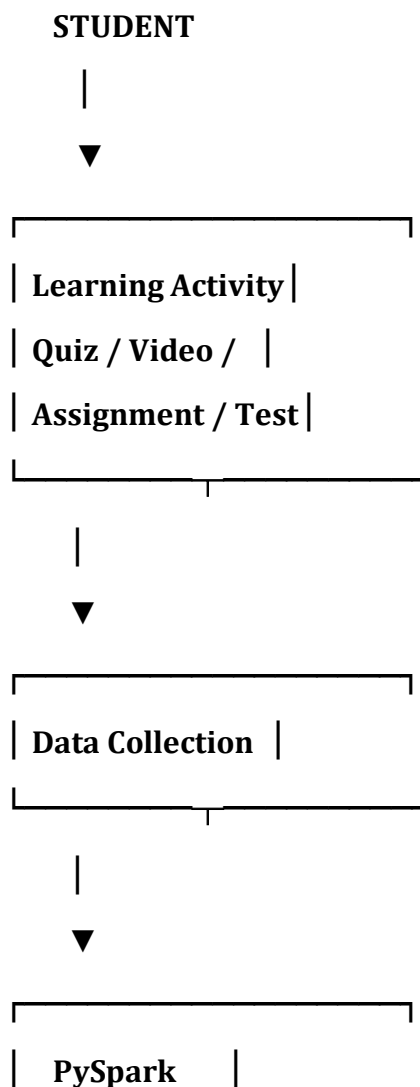
“Design of a Learning Agent for a Personalized Education System Using AI and PySpark”

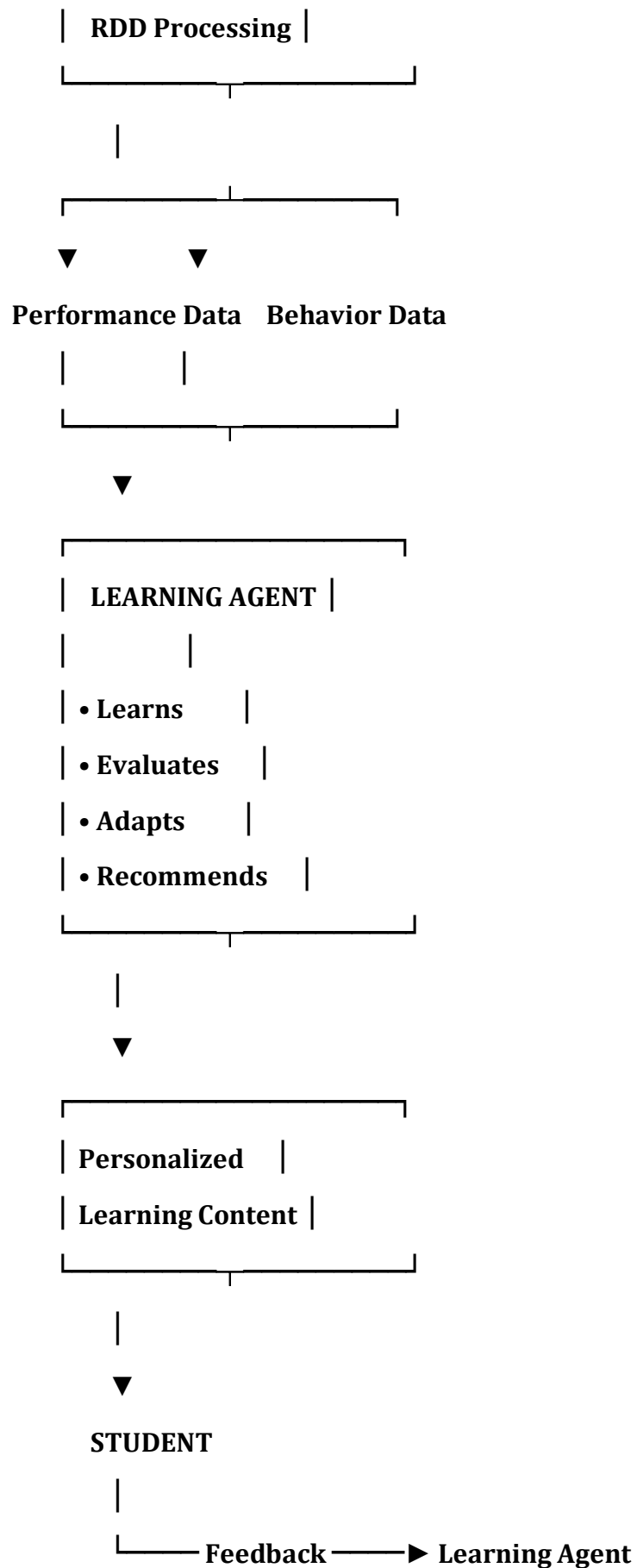
Suggested 10-slide presentation

Slide	Content
1. Title Slide	Learning Agent for Personalized Education System
2. Problem Identification	One-size-fits-all education, different learning speeds, difficulty in monitoring thousands of students
3. What is a Learning Agent?	Agent that learns from feedback and improves its future decisions

Slide	Content
4. System Architecture	Student → Data Collection → PySpark → Learning Agent → Recommendation → Feedback
5. Agent Components	Performance monitor, learning element, critic, problem generator
6. PySpark Integration	RDD creation, map(), filter(), reduceByKey(), groupByKey(), join()
7. Adaptive Learning Logic	Increase/decrease difficulty based on scores, attempts, time and mistakes
8. Real-Time Feedback	Agent receives quiz results and updates the student's learning profile
9. Example + Results	Student performance → agent decision → personalized content
10. Conclusion & Future Scope	Benefits, scalability, AI tutors, predictive learning

System architecture





Learning Agent architecture

The four important components are:

1. Performance Element

- **Decides what learning material should be provided.**
- **Uses the student's current knowledge and performance.**

2. Learning Element

- **Learns from previous student interactions.**
- **Improves future recommendations.**

3. Critic

- **Evaluates whether the recommendation was successful.**
- **Uses quiz scores, completion rate and student feedback.**

4. Problem Generator

- **Selects new learning activities.**
- **Can provide easier, similar or more challenging questions.**

Example decision logic

Suppose a student receives:

- **Quiz score = 45%**
- **Attempts = 3**
- **Wrong answers = 55%**
- **Completion rate = 60%**

The Learning Agent determines that the current topic is difficult.

Score < 50%



Detect learning difficulty



Reduce content difficulty



Provide basic explanation



Give practice questions



Evaluate new result

↓

If score improves → increase difficulty gradually

For a student scoring 90%, the agent can provide advanced questions instead of repeating basic content.

PySpark RDD example

Student data:

Student	Topic	Score
S1	Python	85
S2	Python	45
S3	Python	72
S4	DBMS	90
S5	DBMS	55

RDD creation:

```
data = sc.textFile("student_data.csv")
```

```
rdd = data.map(lambda x: x.split(","))
```

Filter low-performing students:

```
weak_students = rdd.filter(  
    lambda x: float(x[2]) < 50  
)
```

Group students by topic:

```
topic_data = rdd.map(  
    lambda x: (x[1], float(x[2]))  
)
```

```
average_scores = topic_data.groupByKey().mapValues(  
    lambda scores: sum(scores) / len(scores)  
)
```

This demonstrates actual distributed data processing using RDDs.

Example output

Student ID: S2

Topic: Python

Current Score: 45%

Learning Agent Decision:

→ Student needs additional support

→ Difficulty: Reduced

→ Content: Python Fundamentals

→ Practice: 10 basic questions

→ Feedback: Enabled

After Practice:

Score: 78%

Agent Decision:

→ Difficulty increased to Intermediate

Tools you can mention

For the Use of Tools – 20% criterion:

- **Python – agent logic and data processing**
- **PySpark – distributed RDD processing**
- **Jupyter Notebook / Google Colab – implementation and demonstration**
- **Matplotlib – student-performance visualization**
- **CSV dataset – student activity data**

Strong innovation for your presentation

Add an Adaptive Difficulty Score:

Difficulty Score = $0.5 \times \text{Performance} + 0.3 \times \text{Completion Rate} + 0.2 \times \text{Engagement}$

The Learning Agent continuously recalculates this value and changes the difficulty of the next learning activity.

This gives you a clear connection between AI agent → data → decision → feedback → learning, which is excellent for the rubric.

Rubric strategy

Criterion	How to target Level 4
Technical Content – 30%	Explain Learning Agent architecture, components, feedback loop and adaptive decision logic
Problem Identification – 25%	Clearly explain limitations of traditional one-size-fits-all education
Tools – 20%	Demonstrate Python + PySpark RDDs + graphs
Communication – 15%	Use architecture diagrams and one student example
Professionalism – 10%	Mention data privacy, fairness, transparency and responsible AI